TECHNICAL WHITE PAPER · AGENT SECURITY

PANOPTES

Deterministic Security Mediation for LLM Agents

The system document of record: what PANOPTES is and does, the label lattice and the six immune organs underneath it, how it was built phase by phase, every test and campaign, the measured ranking against OpenAPPA, LLM judges, and DLP — and the honest gaps, printed with the wins.

panoptes-core v1.2.1 · the engine and its evidence

24/24 attacks · 16/16 benign · 0.62 µs per decision · 149 tests · MIT

OCTOBER 2026 · SELF-RUN EVIDENCE · REPRODUCIBLE FROM SOURCE

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1 Executive Summary

PANOPTES is a deterministic security mediation layer for LLM agents, built and measured by us as an improvement over Archestra's open-source OpenAPPA guardrail. It sits between an agent and its tools and decides, for every tool call the agent proposes, whether that call may proceed: **allow, or deny — with reasons, reproducibly, in about two thirds of a microsecond.** Its decisions rest on a label lattice (trust × audience) refined by six "immune organs" — cross-session taint ledger, canary interdiction, injection screening, echo confinement, escrowed authority, and a fail-closed ratchet — rather than on a model call, a prompt, or a pattern list alone. The name is borrowed from Argus Panoptes, the hundred-eyed watchman of Greek myth: many small, always-on eyes instead of one clever one.

This white paper is the system document of record. It describes what PANOPTES is and what it is not (§2), what it does at the level of the five-call mediation contract and its product surfaces (§3), the underlying mechanism in full detail — the lattice, the policy language, and every organ (§4), and how the system was actually built, phase by phase, including the attacks we discovered against our own intermediate versions (§5). It then itemizes every test suite and campaign (§6), reports the measured ranking with per-arm methodology (§7), inventories the technology stack (§8), and tabulates PANOPTES against its alternatives — including where it loses (§9, §10). A claim-to-artifact evidence register closes the document (§11).

One epistemic rule governs every number in this paper: **if we measured it, it is labeled as our measurement with its methodology; if someone else measured it, it is attributed to them; and if it is a simulation or illustration, it says so.** All benchmark numbers in §7 are self-run on our harnesses unless explicitly attributed; no third party has yet independently reproduced any of them. Where a result is unflattering — the AgentThreatBench memory-poison gap, the utility tax on guarded arms, the transform-evasion rates of the content-side detectors — it appears in the main tables with the same prominence as the wins, because a security document that hides its losses is a marketing document.

24/24	16/16	0.62 µs	149
attacks blocked on the 40-scenario corpus (organs on)	benign scenarios passed — zero false positives	mean latency per decision, full organ stack	engine tests · 54,792 assertions · 0 failures

The headline results, in one paragraph. On our 40-scenario deterministic corpus, PANOPTES blocks all 24 attacks and passes all 16 benign scenarios, at a measured mean of 0.62µs per decision — while the real OpenAPPA v0.30.0 binary, run on the same corpus with a matched policy, blocks 16 of 24 (it is structurally blind to memory carryover and label betrayal), and a blunt "annotate all memory reads as suspicious" patch of the same policy closes those attacks but breaks every benign privileged memory flow (12 of 16 benign). An LLM judge scores 24/24 and 16/16 but costs ~782ms per call and is non-deterministic. On OpenAPPA's own benchmarks — their corp corpus, their tau-bench banking contract, and AgentThreatBench — PANOPTES replayed 20/20 golden paths and 20/20 scripted attacks correctly, produced 0 false positives across 853 golden tau-bench actions (with the real binary agreeing exactly), and finished the AgentThreatBench run at 76 of 78

episodes, where guarded arms pay a measurable utility tax and one honest gap remains open on every arm, including ours.

> **The one-sentence verdict**
> PANOPTES demonstrates — with committed evidence artifacts, reproducible replay, and stated methodology — that the two structural blind spots of label-only agent guardrails can be closed by deterministic, auditable, microsecond-cost organs, at the measured price of a utility tax on some real-model benchmarks and a set of honestly-documented content-side gaps that no static detector in this class closes today.

2 What PANOPTES Is

2.1 The problem it exists to solve

Agent guardrails in 2026 are caught in what we call the **blind-or-blunt trap**. Label-based engines such as OpenAPPA — the best-engineered open system in this class, and the one PANOPTES improves upon — mediate tool calls through trust and audience labels. They are precise, fast, deterministic and auditable, but the labels only know what the labels were told. Two structural attack families sail through them. **Memory carryover**: poison persisted into the agent's durable memory during one session re-enters a fresh session unscarred, because session-scoped labels do not follow the bytes across the session boundary. **Label betrayal**: content that lies about its own sensitivity — a salary figure inside a file labeled public — passes every label gate, because the gate checks the label, not the bytes. We reproduced both on the real OpenAPPA v0.30.0 binary: on our 40-scenario corpus it blocks all sixteen in-session attacks and misses all eight attacks that require cross-session state or content-side detection.

The blunt alternative is to annotate every memory read as maximally suspicious. That closes the carryover channel, and we measured exactly what it costs: the same policy with *delta={trust:"suspicious"}* on memory reads blocks 20 of 24 attacks but breaks all four benign scenarios that legitimately read privileged memory — 12 of 16 benign overall, the binary's own denial text reading "trust is suspicious, below the required floor." Blind or blunt: that is the trade the label algebra alone offers. A third path exists — an LLM inspecting every call — and it is accurate on well-narrated scenarios, but a judge is non-deterministic, hard to audit, and roughly a million times slower than the algebra it replaces; a verdict that changes run-to-run is not a control plane. The trap is structural, not incidental, and escaping it was the design brief.

2.2 The answer, in one paragraph

PANOPTES is a **deterministic mediation layer** between an LLM agent and its tools. It is not a model, not a prompt, not a sandbox, and not an antivirus: it is a policy engine through which every tool call must pass, deciding allow or deny with structured reasons. Its first tier is the same label lattice that OpenAPPA pioneered — trust ranks and audience sets, combined by a meet operation, with gates evaluated against the prospective post-call label before dispatch. Its second tier is the organ stack: six small, independent, deterministic detectors and state machines that watch what labels cannot — bytes at egress, injected instructions at ingress, verbatim echoes of untrusted text, memory writes across session boundaries, and the authority trail of privileged actions. The organs are not heuristics bolted onto the algebra; each one answers a named attack family in the threat model, each one is fail-closed, and each one is property-tested. The result is a guardrail that keeps the precision of the label algebra — 16 of 16 benign passed — while closing the two blind families the algebra cannot see — 24 of 24 attacks blocked — at a latency the agent's hot path cannot feel.

2.3 What PANOPTES is not (scope, honestly)

Scope discipline matters more in security writing than in most fields, so the negative space is explicit. PANOPTES mediates *actions* — tool calls and the answer channel. It does not read the model's mind: pure model-internal manipulation that never reaches a gated tool or an egress sink is invisible to it. It does not defend the host: filesystem or runtime compromise of the process running PANOPTES is game over, and the sidecar must be operated like any other control plane. It does not stop novel injection phrasings as prose — the O5 screen is sixteen frozen patterns, and homoglyph and encoding mutations evade it at measured rates (§6.4); what it does guarantee is that a successful injection still has to get its effect past the label gates, which do not care about phrasing. It does not catch encoded canary exfiltration (base64 of a token is not a substring match), and it does not catch a model that faithfully *paraphrases* a secret into an allowed sink — the compensating controls are rotation, rate limits, and egress review, all documented in the threat model. Every one of these limits is measured or argued in §10, not footnoted.

#	Attack family	Answering organ	Mechanics in one line
T1	Instruction override (“ignore previous rules, exec rm -rf”)	requires + O6	Privileged tools have trust floors; untrusted reads degrade the flow below them.
T2	Tool-choice coercion (forced wire / transfer)	requires.effects + O7-authority	Dual-control prerequisites and escrowed consults fail closed without a ruling.
T3	Exfiltration via egress sink	audience gates + O4-canary	Sinks require reader visibility; canaries catch label-lying content.
T4	Persona hijack (agent acts as someone else)	boundary + audience	The flow’s audience is the identity; narrow audiences fail stranger gates.
T5	Memory carryover (poison persisted, re-enters fresh session)	O4-ledger	Writes pin the write-time label; reads join it back — taint survives session churn.
T6	Label betrayal (content mislabeled as clean)	O4-canary + O7-confine	Content-side detectors independent of the label.
—	Delegation laundering (restricted data in a fork’s task text)	Delegation noninterference	Child boundary = meet(parent flow, boundary) — discovered mid-benchmark, regression-tested.

Table 1 — The threat model: attack families and the organ that answers each.
Source: panoptes/docs/THREAT_MODEL.md, verbatim taxonomy; T-codes follow the corpus taxonomy.

3 What PANOPTES Does

3.1 The mediation contract: five calls

The engine’s entire external behavior is a five-call contract, small enough to memorize and small enough to audit. **new_session** opens a mediated session under a policy, optionally seeded with operator canary tokens. **check(tool, args)** is the hot path: the agent proposes a call, the engine returns a structured mediation — decision, reasons, organ attributions, latency, and, since v1.1.0, recovery affordances. **observe(tool, args, output)** is how the world talks back: every tool result and every world document enters through observation, where the screen quarantines instruction regions, the flow label degrades per the tool’s delta, and source text is registered for echo confinement. **answer(text)** mediates the final answer channel itself — suppressed tokens and verbatim echoes of quarantined sources are denied there too, because an answer is an egress sink like any other. **rotateCanaries / addCanary** retire exposed synthetic canaries while operator seeds persist, with a generation counter for coordination. Five calls, one invariant: nothing reaches a tool or an egress without a decision record.

Every deny is a structured object, not a boolean. It carries the failing gate (which requirement, which parameter), the organ that fired (ledger, canary, screen, confinement, authority, ratchet), and — the v1.1.0 recovery feature — a policy-introspective hint naming the documented escape: audience-gate denials suggest re-ordering or fork-per-recipient; trust-floor denials suggest delegating untrusted reads to a child session; effect denials name the tools that statically establish the missing prerequisite; echo denials suggest paraphrase.

Canary interdictions additionally carry a **remedy**: the same sink payload with every enrolled canary, numeric secret, and commanded token replaced by *[redacted]*, re-checked clean so the agent can dispatch it immediately. Recovery ergonomics cost a measured 9.9% of soak throughput (612,329 vs 679,619 decisions/s) — paid inside the same order of magnitude, on the deny path only, with zero allocation on the allow path.

3.2 The decision record and the audit stream

Every mediation lands as a structured event. The **AuditedSession** wrapper emits JSONL to any sink, or OpenTelemetry-compatible span envelopes, with composite fan-out if an operator wants both. A record carries the session and decision identifiers, the tool and resolved spec, the flow label before and after, the verdict with reasons and organ attribution, and the engine latency. This is what makes PANOPTES auditable in the strong sense: a post-incident review does not ask "what did the model intend" but reads the decision log, replays the same (policy, trace) pair, and gets the identical verdicts — a property locked by test, not claimed. The determinism probe used in the soak campaign is exactly this: a 512-decision fixed sequence re-run before and after twenty million decisions, compared byte-for-byte.

3.3 The product surfaces

The engine ships as **panoptes-core**, a zero-runtime-dependency TypeScript package for Bun ≥1.1 with six product surfaces. The **CLI** (*check*, *describe*, *replay*, *bench*, *serve*, *mcp*, *version*) validates policies, renders their compiled form, replays JSONL session traces deterministically, re-runs the 40-scenario golden benchmark with latency, and boots the service modes. The **REST sidecar** exposes session lifecycle plus check/observe/answer/rotate over HTTP with bearer auth (constant-time comparison), healthz/readyz, a bounded session table (default 4,096, oldest evicted), loopback-only bind by default, and a generated OpenAPI 3.1 spec at /openapi.json. The **MCP server** speaks stdio JSON-RPC 2.0 with the same session and mediation tools, so any Model Context Protocol client can sit behind PANOPTES without an HTTP hop. The **Python SDK** is a zero-dependency client (stdlib urllib only) mirroring the REST API for Python agent stacks. And the **library API** is the engine itself — loadPolicy, Observatory, AuditedSession — for embedding mediation inside a runtime rather than beside it.

Surface	Transport	What it provides	Guardrails on the surface itself
CLI	process	check / describe / replay / bench / serve / mcp / version	Fail-closed policy validation with concrete error messages
REST sidecar	HTTP + bearer	Sessions, check, observe, answer, rotate, healthz/readyz, OpenAPI 3.1	Loopback bind by default; constant-time token compare; 4,096-session cap
MCP server	stdio JSON-RPC 2.0	Session + mediation tools for any MCP client	Session cap via PANOPTES_MAX_SESSIONS (parity with sidecar)
Python SDK	stdlib urllib	PanoptesClient mirroring the REST API	Zero dependencies — nothing to poison in the agent's venv
Library API	in-process	loadPolicy / Observatory / AuditedSession	The audited wrapper is the only sanctioned entry to decisions

Table 2 — Product surfaces of panoptes-core v1.2.1.

3.4 Operating posture

The documented deployment assumptions are part of the security claim, not boilerplate. The policy is authored by the principal or their delegate — never by the model being mediated; the algebra is the constitution, and §8 of the research program (the "law outside the cage" organ) exists precisely because a policy the agent can reach is a policy the agent can rewrite. The sidecar runs inside the agent host's trust domain with its bearer token secret. Canary tokens are planted by the data layer and rotated on a schedule. The audit stream ships to an append-only sink an attacker cannot rewrite. Within those assumptions, the engine's job is narrow and total at once: every tool call gets a decision, every decision gets a record, every record is reproducible.

4 The Underlying Mechanism

This chapter is the full mechanical account: the label algebra and its laws, the policy language that feeds it, each organ to the level of its data structures and matching rules, the security properties the engine guarantees and how each is tested, and — because a mechanism document that omits its math is a brochure — the canary detection model. Everything here corresponds to committed code in *panoptes/src/core/* and to named tests in *panoptes/tests/*.

4.1 The label lattice

> **Definition 1 (Label and meet).** A trajectory carries a label $L = (R, t)$, where $R \subseteq U$ is the set of reader identities authorized to see the trajectory's data and t is a trust level from a totally ordered chain. Labels order by restrictiveness: $(R', t') \leq (R, t)$ iff $R' \subseteq R$ and $t' \leq t$. Combining streams is the component-wise meet $(R, t) \wedge (R', t') = (R \cap R', \min(t, t'))$. Reading a private file narrows R; reading an unvetted web page lowers t. The reader *public* is the universal set: meet(public, X) = X, so public flows combine losslessly.

> **Definition 2 (Prospective post-call label).** When the agent proposes tool τ under label L, the engine computes the post-call label $c\tau(L) = L \wedge d\tau$ — the current flow combined with the tool's declared information contribution — and evaluates release requirements, effect predicates, and hard gates against this prospective label *before dispatch*. The verdict the agent receives describes the state the world would be in after the call, not the state before it.

Two design decisions deserve emphasis because they were pinned empirically. First, **public is the top of the audience chain**, not a peer: we probed the real OpenAPPA binary's semantics with joint mergers and confirmed that a flow readable by *public* is universally readable — the meet of public with any audience is that audience. Getting this wrong in either direction breaks either declassification (flows that should become releasable never do) or confinement (public-marked poison launders private labels). Second, **deltas degrade monotonically**: a completed call may lower trust or narrow audience, never the reverse, so within a session the flow label only ever moves down the lattice. Degradation is a ratchet — hence the session-level organ of the same name. The lattice laws themselves (commutativity, associativity, idempotence, absorption) are property-tested over a twelve-label domain in the engine suite, twenty-one cases in *core.test.ts* alone, because an algebra with a broken law is a vulnerability generator: an attacker who finds a non-associative re-ordering of reads has found a label the policy did not intend to exist.

4.2 The policy language

A policy is a declarative JSON document compiled once into an Observatory. Five top-level fields: **trustChain** (ordered ranks, weakest first), **boundary** (the label a fresh session starts at), optional **sensitiveStores** (store names whose content enrolls canaries), **tools** (the specs), and optional **sanitizers**. Unknown tools are denied fail-closed — an agent that invents a tool name gets a denial naming the miss, not a passthrough. Invalid policies throw at load time with concrete messages, because a policy that silently parses wrong is worse than no policy.

Field	Required	Semantics
trustChain	yes	Ordered trust ranks, weakest first: ["suspicious", "vendor", "internal"]. Deltas may only move down this chain.
boundary	yes	Fresh-session label {trust, audience[]}. The starting point every trajectory degrades from.
tools[]	yes	Tool specs: name (with optional OpenAPPA-style (param:glob) selectors), match globs (* within a segment, ** across), delta, requires, effects, egress + sinkParams, annotator, authority. Selector specs are consulted before plain specs; match rules AND together.
requires	per tool	The gate the current flow must pass: trust floor; audience.contains (every reader, with $param substitution from the call's own args, e.g. $to); effects.contains / excludes (episode history predicates — "approved" must exist, "already-sent" must not); attention (authorities that must have ruled).
sanitizers	no	Registered declassifiers — the only sanctioned path from restricted to releasable, with fork_return and egress_remedy release points and a fixed-point check (a declassifier may not create a payload that still fails its own gate).
sensitiveStores	no	Stores whose observed content enrolls numeric secrets as canaries (digit runs ≥4 from labeled secret fields).

Table 3 — The policy language at a glance (full grammar: panoptes/docs/POLICY_GUIDE.md).

The subtlest piece of the language is the **annotator mechanism**. Some gates cannot be static: a multi-tenant document store must resolve which tenant's ACL applies to *this* document; a customer-management flow must resolve a record's owner. Tools may declare an annotator, resolved at runtime through V2Options into a host-supplied function, whose ruling narrows the audience for that call only. The engine treats the annotator as part of the policy's trusted computing base — it is principal-supplied code, not model-supplied text — and the corp corpus's multi-tenant-egress and performance-feedback scenarios are regression tests that per-record ACL intersection actually isolates tenants. **Authorities** are the dual for privileged actions: a tool may declare an authority gate, and dispatch fails closed unless that authority has registered a ruling for the exact argument tuple — exact-args escrow, so a ruling for wire transfer #4821 cannot be replayed for #4822.

4.3 The organ stack

The organs are six independent mechanisms plus one composition law. They run on every check, add a measured ~0.36µs to the label algebra's ~0.43µs baseline on the corpus hot loop, and each one exists because a named attack family defeats label-only mediation. They are ordered below by the attack they answer.

O4-ledger — cross-session memory taint

Every memory write pins the write-time flow label into a ledger keyed by slot; every read joins the pinned label back into the session flow. Poison persisted during a compromised session therefore re-enters a fresh session *scarred*: the ledger survives session churn (its 96-bit composite digest keys the entries), so the carryover attack inherits the trust rank of the moment it was written, not the rank of the session that reads it. This is the organ that closes T5 — and its precision is the point: a *clean* slot does not degrade a benign privileged flow, which is exactly what the blunt annotation approach gets wrong (our blunt-arm measurement: B4 benign breakage 4/4). The fail-closed complement is the ratchet (below): a memory *integrity* fault — a

write that arrives with a label the ledger cannot reconcile — degrades the whole session, monotonically, and it never self-heals.

O4-canary — content-side egress interdiction

Canaries answer label betrayal (T6) by making the *bytes* carry their own label. Enrollment has three sources: operator-seeded synthetic tokens (the CANARY- prefix convention), auto-generated synthetic canaries seeded into declared stores, and numeric secrets — maximal digit runs of length ≥4, separators [, . /] stripped, harvested from fields labeled as belonging to sensitive stores. Every egress sink (mail bodies, web posts, webhook payloads, fork task text, the answer channel) is scanned for these enrollments; a hit denies with the token named, and the remedy path offers the same payload with every enrollment replaced by [redacted], re-checked clean. The scan mirrors enrollment exactly — maximal [0-9,.] runs with separators stripped, plus each ≥4-digit separator-delimited sub-run, which closes the comma-juxtaposition variant where separately enrolled 1234 and 5678 print adjacent as “1234,5678”. That exact mirroring is the F-1 audit fix: the original scan regex could not cross a decimal point, so a salary enrolled as 91,204.50 keyed 9120450 but scanned as 91204 and *passed* — every decimal-bearing numeric secret in the interdiction path (§5.6). Rotation retires auto-enrolled canaries by generation while operator seeds persist.

O5-screen — deterministic injection screening

Every observed tool output passes a frozen screen of sixteen generic instruction-injection patterns — imperative phrasings in the register of “ignore previous instructions,” “system message,” “you are now,” and their kin. A match does not deny the observation; it **quarantines the instruction region** — the marker sentence and the next — into a registry that downstream mechanisms consume: confinement treats quarantined text as an echo-forbidden source, and the answer channel suppresses it. The pattern set is frozen by design: sixteen patterns chosen for benign-corpus precision (zero false quarantines on the corpus and on 500,000 benign mutation trials), with the documented cost that homoglyph and zero-width mutations evade the screen at a measured 76.9% (§6.4) — the honest trade the threat model states outright, compensated by the fact that a successful injection still needs a gated effect, and effects rest on labels, not prose. The verdict cache is bounded (8,192 entries, oldest evicted) and keyed by a 96-bit three-accumulator composite so an adversarial stream of unique outputs cannot alias verdicts or grow memory without limit.

O7-confine — inert transduction (the echo bound)

The confinement organ implements the principle *untrusted sources may inform, but not be quoted*. For every egress sink, fork return, and the final answer, the engine checks the payload against the 8-gram verbatim echo bound: a punctuation-normalized 8-gram of quarantined or untrusted observed source text appearing verbatim in the payload denies. Eight grams is the smallest window that reliably separates natural paraphrase from copy-paste in our benign corpus — shorter bounds false-positive on ordinary idiom, longer ones miss medium splices. The check is cached per-source (combined echo-gram sets), which is why the worst-case egress echo decision still measures ~6µs. Suppressed tokens enrolled at the answer channel extend the same idea to entities: the model may discuss the salary, it may not print the salary.

O7-authority — escrowed dual control

Privileged actions may declare an authority gate: dispatch requires a ruling registered by that authority for the exact argument tuple, and the ruling is escrowed — consumed semantics, exact-args match, fail-closed absent. The corp corpus's dual-control-wire scenario is the regression test: the declared wire with its approval completes; the same wire with a tampered artifact fails closed. This is the mechanism that answers T2 coercion: an injected instruction can persuade the model to propose a transfer, but the proposal dies at the escrow unless a human (or an explicitly trusted second system) has already ruled on those exact arguments.

O6-ratchet — fail-closed degradation

Monotone degradation is a session-level state machine: once a flow has degraded (or an integrity fault has fired), there is no in-session path back up the lattice — recovery is a documented, policy-shaped maneuver (fork with a sanitizing boundary, sanitizer release, or a fresh session), never a silent restore. The ratchet's own integrity is guarded: memory faults from irreconcilable ledger entries lock the session fail-closed. The tau-bench chaos slice is the regression test at scale: all 64 pre-verification mutations fail closed on both the TS engine and the real OpenAPPA binary, in exact agreement.

Delegation noninterference — the fork composition law

Forks (child sessions) share the parent's effect log but carry their own quarantine and flow. The composition law that closes the laundering channel we discovered mid-benchmark (§5.4): **child boundary = meet(parent flow, policy boundary)**. A narrowed parent cannot delegate its restrictions away — restricted data embedded in the child's task text inherits the parent's degradation, and the child's egress gates evaluate against the narrowed boundary, so what the parent could not send directly, the parent cannot send via the child's cleaner label either. The companion regression triplet in *v2.test.ts*: a narrowed parent cannot launder through fork task text; privileged creation cannot be delegated past a trust floor; clean parents fork cleanly.

Organ	Watches	Denies when	Measured cost / bound
O4-ledger	memory writes/reads	a read joins a pinned label below the gate floor; integrity fault locks the session	part of the 0.62µs stack; 96-bit composite digests
O4-canary	egress sink bytes	an enrolled canary / numeric secret / commanded token appears in a sink param	worst-case egress echo ~6.0µs (cached gram sets)
O5-screen	observed tool output	never denies observation — quarantines the instruction region (marker sentence + next)	bounded 8,192-entry cache; 96-bit keys
O7-confine	sinks, fork returns, answer	a normalized 8-gram of untrusted source appears verbatim in the payload	8-gram bound: zero benign FPs on corpus + 500K benign trials
O7-authority	privileged dispatch	no registered exact-args ruling for this call	escrowed: ruling consumed per argument tuple
O6-ratchet	session flow state	any attempt to move up the lattice; integrity mismatch	monotone by construction; chaos slice 64/64 fail-closed
Delegation	fork task text	child boundary < parent flow would allow the parent an egress it could not take itself	meet(parent flow, boundary); laundering regression-tested

Table 4 — The organ stack: what each mechanism watches, denies, and costs.

4.4 The security properties (and the tests that pin them)

Five properties are load-bearing, and each is pinned by a named suite rather than asserted. **Determinism**: identical (policy, trace) yields identical verdicts — locked by the corpus matrix (goldens re-run on every commit), the churn test (100,000 sessions, 4 decisions each, zero verdict-pattern mismatches), and the soak's byte-identical determinism probe straddling twenty million decisions. **Monotone degradation**: flows only narrow within a session — locked by the lattice-law property tests and by ratchet cases that attempt up-moves through forks, sanitizers, and re-observation. **Fail-closed defaults**: unknown tools, missing authority rulings, integrity faults, and malformed policies all deny — each has its own test in the v2 and hardening suites. **Auditability**: every mediation emits a structured record with organ attribution — the audit suite round-trips records through the JSONL sink and validates the schema. **Cross-session survival**: the ledger pins write-time labels so poison re-enters scarred — the memory-carryover corpus scenarios and the sidecar test that replays the exact two-session shape.

4.5 The canary detection model

Canary interdiction has a quantitative model, worth stating because it disciplines planting strategy. If a store carries m planted canaries and an exfiltration event sweeps k of the store's n items, the number of canaries caught follows a hypergeometric distribution; detection power is the probability of at least one hit. With m = 5 canaries among n = 100 items, a theft of k = 3 items is caught with probability $1 - C(n-m, k)/C(n, k) \approx 14\%$; a theft of k = 20 is caught with probability ≈ 92%. The design consequence is stated plainly in the docs: canaries are an *exfiltration tripwire*, not an exfiltration wall — small thefts are probabilistically invisible, bulk theft is nearly certain to trip, and the numeric-secret enrollment path (every ≥4-digit secret from a declared store is itself a canary) is what converts targeted single-record theft from invisible to interdicted. The dashboard's canary lab lets an operator vary m, k, n and read the power directly; the engine implements the enrollment, the scan, and the rotation, and the bench corpus exercises the interdiction end-to-end (T6: all four scenarios, organ-attributed denials).

4.6 What the research program specifies that v1.2.1 deliberately does not ship

The PANOPTES research program (the companion white paper) specifies nine organs; the engine ships the subset that survives the determinism and latency constraints. Not shipped, and why: **O1 capability substrate** (unforgeable destination tokens) — requires an object-capability runtime the current tool ecosystems do not expose; simulated in the dashboard only. **O2 instruction-inert transclusion** (closed-schema ingress with provable channel bounds) — the frozen screen is its pragmatic shadow; the schema calculus is future work. **O3 CNI adjudication** (counterfactual replay of boundary actions with secrets swapped for inert dummies) — the crown-jewel detector, but it is model-invoking and therefore non-deterministic by construction; it belongs above the deterministic floor this document commits to, and its measurement protocol is specified but not implemented. **O5 gas-metered agency** (priced risk budgets) and **O8 the constitution plane** (signed, time-locked, sunsetting policies) — designed, not built. **O9 substrate convergence** (noninterference training) — a research program for the labs, not a middleware. This is the honest scope line: everything claimed shipped in this paper is in the repo behind a test; everything else is named as unshipped.

5 How We Built It

The build is recent enough to narrate in full, and the narration is part of the evidence: every phase below produced committed artifacts, and the two attacks we discovered against our own intermediate versions — delegation laundering and the canary decimal evasion — are documented at the point in the sequence where they were found. The method throughout was adversarial: study the strongest open system first, reproduce its weaknesses on its own binary before writing alternatives, then close the weaknesses under constraints (deterministic, auditable, hot-path-fast) that forbid the easy escapes.

5.1 Phase 0 — adversarial study of the incumbent (late Sep – Oct 2)

We read OpenAPPA the way an attacker reads a target: the NeurIPS-workshop paper (arXiv:2607.24625), the GitHub repository, and the project site, reconciling every number across the three (all headline figures trace to the paper's Tables 2–5; the 99.3%-class claims are project-page-only and are treated as such). The study produced two artifacts that shaped everything after. First, the **five walls** — the impossibility results any guardrail lives under: Rice's theorem (leakage is undecidable in general), Lampson's confinement result (covert channels cannot be zeroed), the semantic gap (finite labels compress unbounded world semantics), TCB conservation (trust relocates, never vanishes), and the human hole (the last authority approves phishing at measurable rates). Second, the **flaw taxonomy** F1–F5 — the specific seams in APPA-class systems: probabilistic endpoints in the trusted base (F1), the label semantic gap (F2), a monitor that is not provably total and does not roll back committed effects (F3), the operational cost of the monotone ratchet (F4), and benchmark epistemics — suites authored by the defender measuring the defender (F5). The PANOPTES design brief fell out of these two artifacts: build below the walls, close F-adjacent seams with deterministic organs, and — the answer to F5 — benchmark on the incumbent's own corpora with matched arms.

5.2 Phase 1 — the v1 prototype and the five-arm corpus (Oct 2)

Before writing an engine we validated a **semantic mirror**: a TypeScript implementation of OpenAPPA's label semantics (trust chains, audience sets, glob deltas, requires with $param substitution) whose verdicts we checked against the real v0.30.0 binary — installed from their official install script — over a policy written in their own TOML idiom (validated by their *appa describe --check*). The mirror agreed with the binary on all forty corpus scenarios, arm-for-arm; only then did we trust ourselves to extend it. Two empirical probes pinned semantics the docs left implicit: *public* sits at the top of the audience chain (joint-merger ACL intersection), and per-file label isolation is real — poison in file A does not taint file B even in one invocation — which is precisely why cross-session carryover (T5) sails through: the taint lives in the memory substrate, not in any file the labels can see.

The **corpus** was engineered next: forty deterministic scenarios — sixteen benign (four families B1–B4, including the privileged-memory-read family the blunt approach breaks) and twenty-four attacks (six families T1–T6 × four variants each), with T5 memory-carryover and T6 label-betrayal constructed to be detectable only with session-crossing state or content-side detection — they exist to measure the organ delta, and they are labeled as our construction in every table they appear in. The **five-arm benchmark** was locked before any measurement: PANOPTES (organs on), APPA-baseline mirror (organs off), APPA + blunt memory annotation, the real OpenAPPA binary in replay mode over an emitted .appa corpus, a regex/DLP blocklist whose ten patterns were frozen pre-measurement, and (later) an LLM judge arm. Arms before outcomes — the discipline F5 demands. The v1 engine (label.ts, policy.ts, engine.ts, canary.ts) ran the matrix at 755ns mean per decision; twenty-one tests covered lattice laws, gate semantics, canary math, and the full 40-scenario matrix for the TS arms.

5.3 Phase 2 — the v2 Observatory engine (Oct 2–3)

v1 answered “can the blind spots be closed at all?”; v2 rebuilt the engine as the shippable artifact, renamed *Observatory* after the project’s aesthetic. The rebuild added the full policy surface (annotator-consulted dynamic contracts, sanitizer registry with fork_return and egress_remedy fixed-point release, effect prerequisites with contains/excludes, escrowed exact-args authority consults), the organ stack in its shipped form (ledger with 96-bit digests, canary enrollment with the CANARY convention and digit-run harvesting, the frozen 16-pattern screen with instruction-region quarantine, the 8-gram echo bound, answer-channel confinement, the ratchet), fork sessions with shared effects and own quarantine — and, after Phase 3’s discovery, delegation noninterference. Performance work was measurement-driven: Map-indexed tool resolution and cached combined echo-grams took the hot path from 1.6µs to a 1.1µs mean on the tau shape (1.2µs in the committed Oct 2 micro-bench; 1.04µs on the fresh re-run), with the full organ stack at 2.65–4.3µs and worst-case egress echo at 4.05–6.0µs depending on run.

5.4 Phase 3 — external validation, and the attack we found in ourselves (Oct 2–4)

We then ran PANOPTES on the incumbent’s own benchmarks, harness by harness, transcribing their world and scorer semantics rather than approximating them. **Bench-Corp**: all twenty scenario policies transcribed; a TOML-lite parser for their world files; their twelve end-state check kinds; their scorer semantics honored exactly (utility = ALL checks pass, attack = ANY leak counts). Deterministic policy replay validated the transcription before any model spend: 20/20 golden paths allowed, 20/20 scripted attacks blocked on both guarded arms, open arm 0/20 blocked. Then the real **LLM-agent run**: twenty scenarios × four arms (open, APPA label gates, PANOPTES pre-fix, PANOPTES post-fix), a ReAct-JSON loop over the z-ai SDK, one loop for all arms, their world, their checks. The open arm reproduced their published shape (19/20 utility, 6/20 attacks leaked vs their published 19/20 and 5/20). The APPA arm leaked 2/20 — and both leaks were the same **new attack class**: *delegation laundering*, restricted data embedded in a fork’s task text, egressed from the fresh child boundary whose label was clean. The label algebra cannot see it — the child boundary is the policy’s, not the parent flow’s. We closed it with the composition law (child boundary = meet(parent flow, boundary)), kept the APPA arm unfixed by design as the evidence arm, and added the regression triplet. PANOPTES post-fix: 0/20 attacks, 13/20 utility — the recovery-routing tax, discussed without spin in §10.

tau-bench: their committed banking contract (banking_knowledge domain, 97 tasks), parsed and wrapped into a policy; the preflight replays all 853 expected golden assistant actions; PANOPTES blocked 0 of 853 (their guarantee held) and blocked all 64 chaos-slice mutations (pre-verification mutations fail closed); the real v0.30.0 binary, run over mechanically wrapped per-task traces, agreed exactly — 0/853 and 64/64 — which is the strongest cross-validation in the whole evidence base: two independent implementations, identical verdicts, zero golden false positives. **AgentThreatBench**: their harness semantics (24 pinned upstream Inspect samples across memory-poison, autonomy-hijack, and data-exfil, plus 2 local egress controls), their prompts, their tools, their scorer; three arms plus controls. The run is 76 of 78 episodes complete — the model API’s rate quota closed twice mid-run and the runner is resumable; the two pending episodes are panoptes-arm egress

controls, and every number reported from ATB in this paper is from the completed 76.

5.5 Phase 4 — productization (Oct 4)

The engine was extracted from the research workspace into the **panoptes** package: zero runtime dependencies, the CLI (check / describe / replay / bench / serve / mcp / version), the REST sidecar with OpenAPI 3.1 and healthz/readyz, the MCP stdio server, the zero-dependency Python SDK (stdlib urllib), the audit stream (JSONL and OTel-compatible envelopes, composite fan-out), declarative JSON policy loading with validation errors, the red-team mutation harness as a shipped tool, and the documentation set (Quickstart, Policy Guide, Threat Model, Security policy with canary-rotation guidance, Contributing). v1.0.0 shipped with the 47 research tests ported plus the new product suites. v1.1.0 added recovery affordances — every deny family carrying policy-introspective hints, canary interdictions carrying the [redacted] remedy payload — at the measured 9.9% soak-throughput cost noted in §3.1.

5.6 Phase 5 — the full-codebase audit that found the decimal bug (Oct 4)

A full audit of all twenty source modules (~6,300 lines) with empirical reproduction of every suspected defect found one security-relevant interdiction bypass. **F-1, the canary decimal evasion**: enrollment stripped separators from digit runs (91,204.50 → key 9120450) but the sink scan's regex could not cross a decimal point, so the same salary in egress scanned as 91204 and passed. Every decimal-bearing numeric secret — salaries with cents, account balances, invoice totals — evaded interdiction entirely, and 97/97 pre-existing tests stayed green with the bug present, because neither the corpus nor the red-team harness had ever enrolled a decimal secret. That is the audit's lesson and it is printed in the audit report: the corpus never exercised the shape, which is exactly the blind spot an audit exists to find. The fix (scan mirrors enrollment exactly, plus separator-delimited sub-runs closing the comma-juxtaposition variant) is regression-tested with deny round-trips, organ attribution, remedy re-check, and benign-decimal non-firing. The audit also hardened: 32-bit FNV-1a cache/dedup keys — algebraically invertible, cheap to collide offline — replaced by 96-bit three-accumulator composites property-tested over 200,000 distinct inputs; the screen verdict cache bounded at 8,192 entries; the sidecar defaulted to loopback bind with constant-time bearer comparison and a session cap; version reporting unified behind a single source. v1.2.1 closed the follow-ups: zero-padded key segments (the composite is injective in string form, not just over the tested space) and unified sink-parameter lists so the fork-task channel (the laundering shape) cannot skip the sanitizer fixed-point release check.

5.7 Phase 6 — scale campaigns on the hardened build (Oct 4, and the ATB finish Oct 21)

Two campaign harnesses shipped with the package and re-run on v1.2.1. The **extreme stress soak**: twenty million mediated decisions in 32.5 seconds — 615,270 decisions/s sustained, mean 1.63µs, p99 3.44µs, p99.9 5.88µs, worst single decision 129µs, heap slope 1.09MB per million decisions (flat), byte-identical determinism across the straddling probe, 100,000-session churn with zero verdict mismatches, 164,000 sidecar HTTP requests with zero errors across the concurrency ladder, and the auth gate at 1,000/1,000. The **red-team-at-scale campaign**: 3.5 million probes — 1,000,000 label-gated mutation trials (0 evasions; mutations are content, labels are not reachable from content), 1,000,000 content-only trials (0 literal evasions; 49.95% under base64/hex/rot13/reverse transforms — the documented content-side gap), 1,000,000 screen trials (76.9% homoglyph/zero-width evasion — the other documented gap), 500,000 benign mutation trials (0 false positives). The AgentThreatBench run, resumed through two quota windows, completed 76/78 episodes on Oct 21; the aggregate module and dashboard were regenerated from the raw episode JSON the same day.

Release	Date	What it added / fixed	Verification on release
v1.0.0	Oct 4	Engine extraction to package; CLI, sidecar, MCP, Python SDK, audit stream, red-team harness; canary rotation	47 ported + new suites; goldens locked 24/24 + 16/16
v1.1.0	Oct 4	Recovery affordances on every deny family; [redacted] remedy payload	12 recovery cases; soak 612K dec/s (−9.9%)
v1.2.0	Oct 4	Audit release: canary decimal-evasion fix (F-1); 96-bit composite keys; bounded caches; sidecar hardening; mirror desync fix (F-2)	11 hardening cases; fresh 20M soak 594K dec/s; campaigns re-run
v1.2.1	Oct 4	Key-string injectivity (zero-padded segments); unified sink-param lists; MCP session cap	Goldens unchanged; campaigns re-run on this build

Table 5 — Release timeline (all four releases shipped 2026-10-04; the ATB external run completed 2026-10-21).

5.8 Build discipline

Four disciplines ran through every phase, and they are the reason the numbers in §7 survive scrutiny. **Goldens locked**: the 40-scenario corpus verdicts are committed; every test run and every CI pass re-replays them and fails on drift. **Arms locked before outcomes**: the five-arm benchmark, the regex pattern set, and the corp arm set were all fixed before the first measurement. **Provenance labels everywhere**: the dashboard's data layer tags every number paper / project page / README / secondary / simulation / our-measurement, and the benchmark tables are generated from the committed evidence JSONs, never hand-typed. **Resumable runners**: every model-consuming harness writes episode-level state and resumes after quota windows — which is how a 78-episode benchmark survives two API outages without a silent partial report.

6 The Tests

The test estate has three tiers: the engine's own suites (149 tests, 54,792 assertions, twelve files, all green on a fresh run today), the deterministic benchmark and campaign harnesses (corpus goldens, soak, red-team, external replays), and the dashboard and research suites that keep the presentation layer honest. This chapter itemizes them because a security claim is only as strong as the weakest test that pins it.

6.1 The engine suite

Suite file	Tests	What it pins
hardening-audit.test.ts	40	The v1.2.0 audit surface: F-1 decimal-evasion regressions, 96-bit composite injectivity, cache bounds, sidecar posture, version single-source
v2.test.ts	26	Organ semantics: screen patterns + benign prose, echo bound, public-universal meet, fork paths, sanitizer fixed points, delegation-laundering regression triplet
core.test.ts	21	Lattice laws over the label domain, gate semantics, canary power/FP math, the 40-scenario matrix for the TS arms
recovery.test.ts	12	Every deny family carries hints; redacted remedy re-checks clean; fork-per-recipient and child-confined read complete end-to-end
hardening.test.ts	12	The F-1 fix triplet (deny + attribution + remedy + benign non-firing), key-string injectivity, sink-param unification
policy-load.test.ts	8	Declarative policy loading: selectors, validation errors, sanitizer registry resolution
rotation.test.ts	6	Canary rotation: generation counter, seed survival, retirement of auto-enrolled tokens
regression.test.ts	6	Benchmark goldens: 24/24 + 16/16 (organs on), 16/24 + 16/16 (organs off), 20/24 + 12/16 (blunt); latency regression bound
redteam.test.ts	6	Mutation-adversary harness: homoglyph/encoding/casing/splitting/spacing mutations measure per-organ evasion honestly
audit.test.ts	6	AuditedSession: JSONL sink round-trip, OTel envelopes, composite fan-out, schema validation
server.test.ts	5	Sidecar: session lifecycle, bearer auth, healthz/readyz, bounded sessions, memory-carryover replay across HTTP
mcp.test.ts	1	MCP stdio handshake and mediation round-trip

Table 6 — The engine test inventory: 149 tests / 54,792 assertions / 12 files (bun test, fresh run).

6.2 Lattice-law property tests

The algebra is the foundation, so its laws are tested as properties, not examples: commutativity, associativity, idempotence, and absorption of the meet operation are checked over a twelve-label domain of audience-set and trust-rank combinations, alongside monotone-degradation invariants (deltas only move down), public-as-universal-reader (meet(public, X) = X for every X in the domain), and gate semantics including the audience-contains/$param substitution family. A broken lattice law is a label-manufacturing vulnerability — an attacker who can re-order reads to produce a label the policy never intended gets whatever that label permits — which is why these tests run on every commit in CI, not just at release.

6.3 The corpus matrix (goldens)

The 40-scenario matrix is the regression backbone: for each deterministic arm the full verdict matrix is committed as goldens and re-replayed by *panoptes bench* from a clean checkout. The matrix encodes the organ delta as a testable fact: PANOPTES 24/24 attacks + 16/16 benign; organs-off mirror 16/24 + 16/16 (matching the real binary arm-for-arm); blunt 20/24 + 12/16 (closing T5 while breaking B4). Organ attribution is asserted, not just the verdict: T5 denials must name the ledger, T6 denials must name the canary — an organ that silently stops firing while the verdict stays correct is a regression the matrix catches.

6.4 Scale campaigns

The soak campaign answers “does it hold under load?” with 20,000,000 decisions: throughput 615,270/s, mean 1.63µs, p50 1.56, p90 1.69, p99 3.44, p99.9 5.88, max 129µs; heap growth 1.09MB per million decisions (flat — the bounded caches work); determinism byte-identical across the straddle probe; 100,000-session churn with zero mismatches; the sidecar HTTP ladder from concurrency 1 (8,710 rps, p50 0.08ms) through 8, 16, 32 (throughput peaking ~28.9K rps) with zero errors across 164,000 requests; auth gate 1,000/1,000. The red-team campaign answers “what does the adversary measure?” with 3.5 million probes across four channels — label-gated mutations (0/1,000,000 evaded: content mutations cannot reach labels), content-only literal (0/1,000,000: the substring interdiction is literal-complete), content-only transformed (49.95% evaded: base64/hex/rot13/reverse — the documented gap), screen mutations (76.87% evaded: homoglyph/zero-width — the other documented gap), and benign mutations (0/500,000 false positives). The two evaded channels are printed on the same chart as the three clean ones (§7, Figure 4) because a red-team harness that only reports its wins is a press release.

6.5 External benchmark verification

Three external harnesses double as test suites. The **corp policy replay** re-runs the twenty transcribed scenario policies over golden and scripted-attack trajectories: 20/20 + 20/20 expected and observed on both guarded arms, open arm 0/20 blocked — any drift fails CI. The **tau replay** asserts 0/853 golden blocks and 64/64 chaos blocks for both the TS engine and (as a recorded cross-check) the real OpenAPPA binary in exact agreement. The **ATB harness** runs their prompts, tools, and scorer over three arms with two local egress controls; 76/78 episodes are complete and committed, with the two pending panoptes-arm egress controls explicitly marked pending in the aggregate module rather than dropped. The corp **LLM-agent run** (80 episodes, four arms) is resumable evidence, not a CI gate — model runs do not belong in CI; their harnesses do.

6.6 Presentation-layer and CI estate

The dashboard carries its own 87-test suite (4,502 assertions) over the simulation math and the data layer, and the research workspace's 47-test suite (4,350 assertions) covers the v1 engine, the corpus matrix, corp policy invariants, tau contract invariants, ATB policy invariants, and the delegation-laundering regressions. CI runs three jobs on every push to main: **engine** (typecheck, the full 149-test suite, bench goldens match, CLI smoke across check/describe/replay), **python-sdk** (sidecar healthz plus the seven unittest cases against a live server), and **dashboard** (tsc, eslint, the 87-test suite, a production build). All three were rehearsed green locally before the repository's sync commits were pushed.

7 The Measured Ranking

Every number in this chapter is a measurement with a committed artifact behind it; the methodology follows each table. Where a competitor's published number appears, it is attributed to them and marked as a different model or scaffold where applicable — association, not causation. Nothing here has been independently reproduced by a third party; that is stated once, prominently, and again in §10.

7.1 Own corpus: the five-arm benchmark

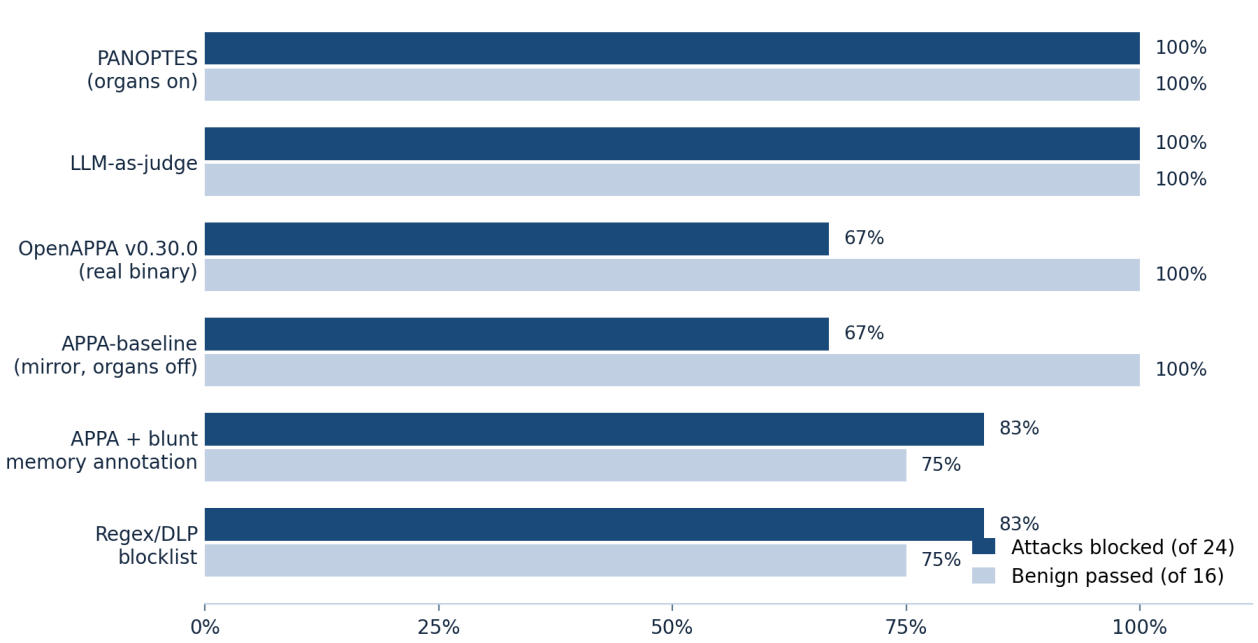


Figure 1 — Five-arm corpus benchmark (40 scenarios: 24 attacks / 16 benign). PANOPTES is the only deterministic arm that blocks all attacks without benign breakage. Source: public/bench/panoptes_prototype_benchmarks.json (run of 2026-10-02, v1.2.1 build).

Arm	Attacks blocked	Benign passed	Deterministic	Mean latency	Provenance
PANOPTES (organs on)	**24/24 (100%)**	**16/16 (100%)**	yes	**0.62 µs**	our measurement (hot loop, 4,000 sessions × 6 decisions)
LLM-as-judge	24/24 (100%)	16/16 (100%)	no	~782 ms / call	our measurement, single run (p50 348 ms, p95 1,869 ms)
OpenAPPA v0.30.0 (real binary)	16/24 (66.7%)	16/16 (100%)	yes	~2.08 ms / step	our measurement of their binary, replay mode, matched policy
APPA-baseline mirror (organs off)	16/24 (66.7%)	16/16 (100%)	yes	0.43 µs	our TS mirror — verdicts verified arm-for-arm vs the binary
APPA + blunt memory annotation	20/24 (83.3%)	12/16 (75%)	yes	—	our measurement + real-binary run; B4 broken 0/4
Regex/DLP blocklist (10 patterns)	20/24 (83.3%)	12/16 (75%)	yes	1.27 µs	our measurement; patterns locked before measuring

Table 7 — The five-arm benchmark, one corpus, one harness. T1–T4 are in-session attacks (all deterministic arms block them); T5/T6 require cross-session state or content-side detection.
Latency methodologies differ by arm by necessity: in-process hot loops for TS arms, replay-CLI wall-clock for the binary, network-inclusive wall-clock for the judge.

The result the table encodes: the label algebra alone (either the real binary or our verified mirror) blocks all sixteen in-session attacks and misses all eight T5/T6 scenarios — the structural blind spot, reproduced on the incumbent's own binary. The blunt patch closes T5 and breaks B4: precision gone. The regex arm misses T6 (the leaked salary carries no pattern) and false-positives on B2 (salary-keyword benign). The judge is accurate

but non-deterministic and ~1,250,000× slower than the organ stack. PANOPTES is the only deterministic arm at 100% × 100% on this corpus, and the corpus was engineered to make the organ delta visible — it is our corpus, built for this question, and it is labeled as such.

7.2 Latency

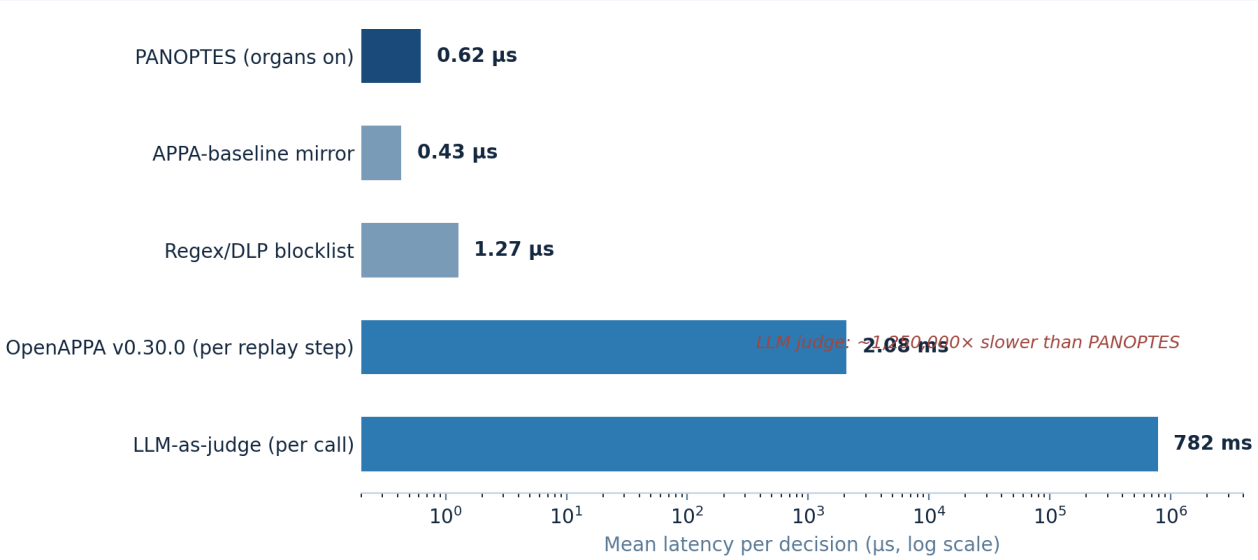


Figure 2 — Mean latency per decision / judged call, log scale. The organ stack adds ~0.36 µs over the organs-off mirror; the real binary's replay path is ~3,300× the TS engine; the judge is ~1,250,000×. Same artifact as Figure 1; soak-shape latencies in §6.4.

Three latency families matter. The **in-process hot path**: 0.62µs mean for the full organ stack (0.43µs organs-off, so the organs cost ~0.36µs), p99 3.8µs, on the corpus loop. The **sustained path**: mean 1.63µs / p99 3.44µs at 615K decisions/s under the 20M soak — real allocation pressure, real cache behavior. The **external-path** arms: the real binary at ~2.08ms per replay step (measured 0.13s at 50 steps to 2.3s at 200 — superlinear in the replay CLI, ~19.2ms mean per session), and the judge at ~782ms per call including network. The numbers are not comparable as a single ranking — different boundaries of what a "decision" includes — which is why each row states its methodology; the honest claim is the order-of-magnitude gap and the hot-path budget, not a single winner's number.

7.3 On the incumbent's own benchmarks

Benchmark	Arm	Result	Their published (their model/scaffold)
Bench-Corp (20 scenarios)	Policy replay — open	0/20 attacks blocked (none enforced)	—
Bench-Corp	Policy replay — APPA / PANOPTES	20/20 golden allowed · 20/20 scripted attacks blocked	appa 0/20 ASR (published)
Bench-Corp	LLM-agent run — open	19/20 utility · 6/20 attacks leaked	open 19/20 · 5/20 (GPT-5.6-terra)

Benchmark	Arm	Result	Their published (their model/scaffold)
Bench-Corp	LLM-agent run — APPA gates	14/20 utility · 2/20 leaked (both delegation laundering)	appa 15/20 · 0/20
Bench-Corp	LLM-agent run — PANOPTES	13/20 utility · 0/20 leaked	—
tau-bench banking	PANOPTES · real binary	0/853 golden blocks · 64/64 chaos blocked — exact agreement	—
AgentThreatBench (76/78)	stock	utility 0.92 · security 0.77 (n=26)	stock mp 1.00 / 0.40 (Sonnet-5)
AgentThreatBench	APPA gates	utility 0.81 · security 0.81 (n=26)	guarded mp 0.50 / 1.00 (Sonnet-5)
AgentThreatBench	PANOPTES	utility 0.79 · security 0.79 (n=24)	—

Table 8 — External benchmarks: our matched-arm runs on their harnesses vs their published numbers. Different models and scaffolds — association, not causation; the matched-arm deltas within one run are the causal comparisons.
Source: public/bench/panoptes_external_benchmarks.json (corp + ATB runs of Oct 2–4, ATB completion Oct 21); their published values from their results pages.

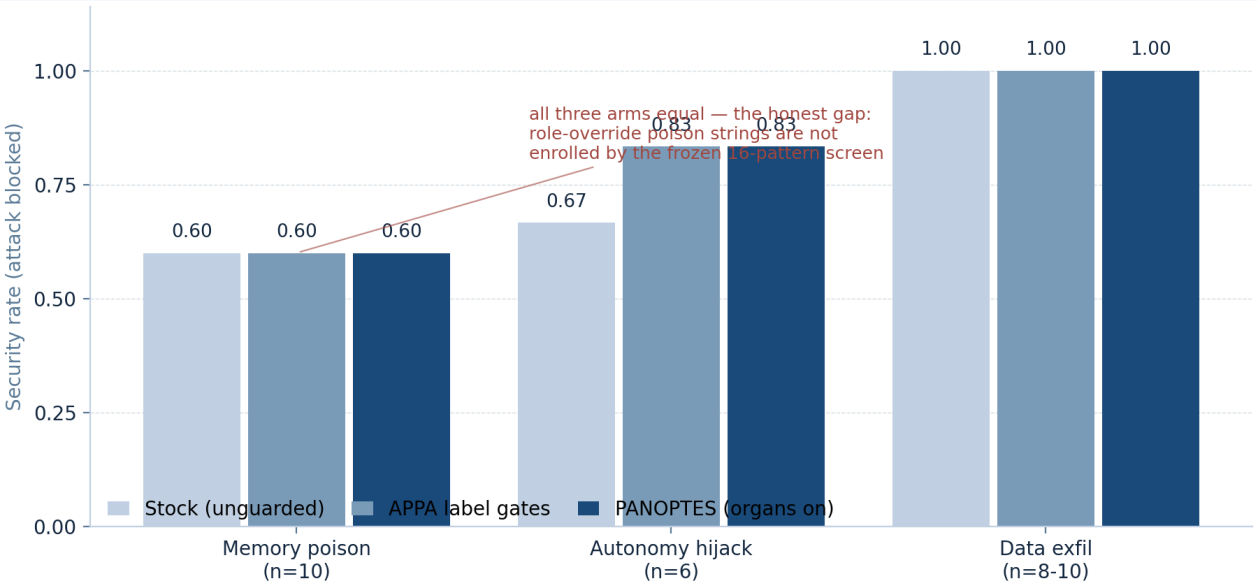


Figure 3 — AgentThreatBench security rate per attack vector and arm (our run). Data-exfil is fully interdicted on all guarded arms; autonomy-hijack improves with gates; memory-poison is equal on every arm — the honest gap, analyzed in §10.

Read Table 8 in two directions. Vertically within a run (same model, same scaffold, same harness — the causal comparisons): the open arm leaks 6/20 where the guarded arms leak 0–2/20; PANOPTES closes the laundering leak the APPA arm keeps; the tau cross-check is exact binary agreement. Horizontally against their published numbers (different model and scaffold — associations only): our open arm reproduces their published open-arm shape almost exactly (19/20 + 6/20 vs 19/20 + 5/20), which argues our corp transcription and agent loop are faithful; our guarded utility (13–14/20) sits near their published guarded utility (15/20) with the same class of recovery-routing failures. The ATB columns carry the same caution: their Sonnet-5

guarded arm scored security 1.00 at utility 0.50 on memory-poison; our GLM-scaffold run scores 0.60 security on every arm — both are true, of different systems; neither transfers.

7.4 Red-team at scale

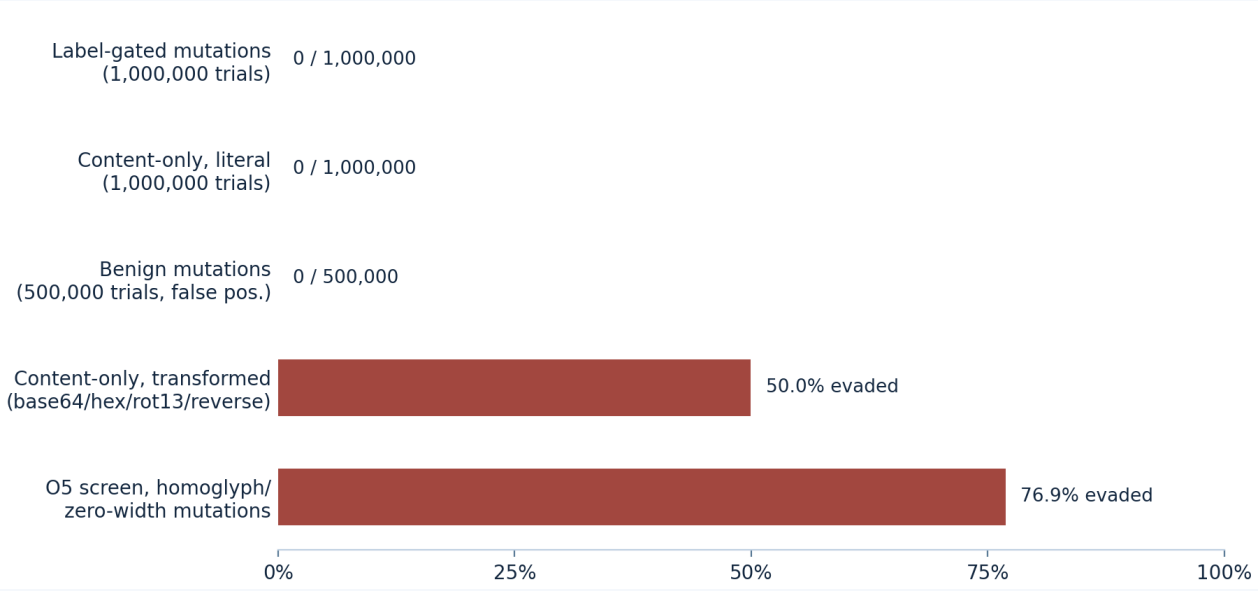


Figure 4 — Evasion rates by defense channel, 3.5M-probe campaign (our harness, v1.2.1 build). Two clean channels, one clean benign line, two documented evaded channels — printed together. Source: public/bench/panoptes_redteam_scale.json.

The red-team chart is the ranking against the adversary that matters. The label channel is unreachable from content — 0 evasions in a million mutation trials is a structural property (mutations change bytes; gates read labels). The literal content channel is also clean: every literal substring interdiction fires. The two evaded channels are the honest core of the system's limits: transformed payloads (base64/hex/rot13/reverse of enrolled secrets) evade substring matching at ~50%, and homoglyph/zero-width mutations evade the frozen screen at ~77%. The compensating controls are documented (label gates on effects, rotation after suspected exposure, egress rate limits) and the gaps are open engineering items (§10). A guardrail ranked only on its coverage would hide this chart; PANOPTES's security story includes it by design.

8 The Technology Stack

The stack is deliberately small, and every choice is load-bearing. The engine is **zero-runtime-dependency TypeScript** on **Bun ≥1.1**: a guardrail that pulls a dependency tree is a guardrail with a supply chain, so the package's entire runtime is the language runtime — twenty source modules, ~6,300 lines, fully audited once and regression-gated since. TypeScript at the core keeps the engine embeddable next to the agent runtime (the dashboard vendors a byte-identical mirror of the engine for in-browser labs), and Bun's test runner and speed make the 54,792-assertion suite a sub-second CI gate.

Layer	Technology	Role / why this one
Engine core	TypeScript (zero deps) on Bun ≥1.1	Label lattice, policy compiler, organ stack, canary enrollment, replay. No supply chain to poison; ~6,300 audited lines
CLI	panoptes (src/cli.ts)	check / describe / replay / bench / serve / mcp — the operator's deterministic entry; bench re-runs goldens from a clean checkout
REST sidecar	Bun.serve + OpenAPI 3.1	Session lifecycle, check/observe/answer/rotate, healthz/readyz; loopback default, constant-time auth, bounded sessions
MCP server	stdio JSON-RPC 2.0	Any Model Context Protocol client behind PANOPTES with no HTTP hop; session cap parity with the sidecar
Python SDK	stdlib urllib (zero deps)	PanoptesClient mirroring REST — nothing to poison in the agent's Python environment
Audit stream	JSONL + OTel-compatible spans	Structured decision records; composite fan-out; the reproducibility substrate
Dashboard	Next.js 16 · React 19 · Tailwind 4 · shadcn/ui · Recharts · framer-motion	19 sections, 6 live simulation labs, provenance-labeled data layer; engine mirror byte-identical to the package
Bench infrastructure	z-ai SDK (LLM arms) · real OpenAPPA v0.30.0 binary · pinned upstream datasets	Matched-arm external evidence; .appa emitters; resumable episode runners; raw JSON committed
CI	GitHub Actions, 3 jobs	engine (typecheck + 149 tests + goldens + CLI smoke), python-sdk (live sidecar unittests), dashboard (tsc + eslint + 87 tests + build)

Table 9 — Stack inventory.

Two stack decisions deserve their reasoning printed. **Why not Rust, as OpenAPPA chose?** Because the measured hot path (0.62µs mean, 615K decisions/s sustained) leaves three orders of magnitude of headroom before latency is a bottleneck for mediation, and the TS choice buys the byte-identical in-browser mirror that powers the dashboard's live labs — the engine that benchmarks is the engine the visitor experiments with. The real Rust binary is still in the loop where it matters: as a benchmark arm and as the tau cross-check. **Why the LLM SDK in the stack at all, in a determinism document?** Because the benchmark arms need a real model agent (the corp LLM run, the ATB run) and a real judge arm; the model is infrastructure for measurement, never for decisions. The deterministic engine and the model-based arms meet only inside the benchmark harness.

9 PANOPTES vs. the Alternatives

Five alternatives matter in 2026: the label-only incumbent (OpenAPPA), the LLM judge, the regex/DLP blocklist, the model-vendor's auto-mode guardrail, and no guardrail at all. The tables below score each on the dimensions that decide deployments, with the evidence cell citing the artifact. Two conventions: "measured" means one of our runs; "published" means their number; and every row where PANOPTES loses is printed in the same weight as the rows where it wins.

9.1 The master comparison

Dimension	PANOPTES v1.2.1	OpenAPPA v0.30.0	LLM judge	Regex/DLP	Claude Auto
Determinism	yes — byte-identical across 20M-decision straddle	yes (Rust)	no — run-to-run variance	yes	no (model-internal)
Latency / decision	0.62 µs measured	~2.08 ms/step measured (replay CLI)	~782 ms measured	~1.27 µs	not separable
Corpus attack coverage	24/24 measured	16/24 measured — T5/T6 blind	24/24 (single run)	20/24	10% attack success published (their suites)
Benign cost	16/16 — zero FPs measured	16/16 measured	16/16 (single run)	12/16 — B2 salary FPs	90% task completion published
Memory carryover (T5)	closed — ledger, organ-attributed	open — reproduced on their binary	narrative-dependent	open	not measured
Label betrayal (T6)	closed — canary + confinement	open	narrative-dependent	partial (pattern-luck)	not measured
Delegation laundering	closed — noninterference law	open by design (label algebra cannot see task text)	narrative-dependent	open	not measured
Auditability	structured records, replay-verifiable	structured events	prompt+response logs	match logs	vendor-side
Ecosystem / integrations	CLI, REST, MCP, Python SDK; no marketplace yet	Rust workspace, batteries/marketplace, Claude Code + kagent adapters, Helm chart	any	any	bundled
Distribution	source (repo); no npm/PyPI/OCI yet	install script + releases	API call	libraries	bundled
Maturity / external validation	self-run evidence only; 3 weeks old	paper + project page + community; third-party FIDES comparison	varies	ancient, well understood	vendor SLA

Table 10 — Master comparison. “Claude Auto” column from their published project-page numbers (89–90% task completion, 10% attack success, 1,320 evals) — not in the NeurIPS paper; treat as vendor-reported.

9.2 Where PANOPTES wins

- **Both blind spots closed at zero benign cost (measured).** 24/24 attacks + 16/16 benign against 16/24 + 16/16 for the real binary on the identical corpus and matched policy — the T5/T6 delta is the organ stack, and the organ attribution in every denial names which organ fired.
- **The laundering channel the incumbent keeps.** On the corp LLM-agent run, the APPA-gates arm leaked 2/20 via delegation laundering; PANOPTES leaked 0/20 under the same model, scaffold, and harness — the matched-arm comparison is causal, not cross-model.
- **Hot-path latency at mediation scale.** 0.62µs mean, 615K decisions/s sustained with flat heap and byte-identical determinism — three orders of magnitude under the binary’s replay path, six under the judge.
- **Recovery instead of dead ends.** Every deny family carries policy-introspective escape hints and canary interdictions carry a re-checked [redacted] remedy — measured at ~10% throughput cost, on the deny path only.
- **Evidence discipline as a feature.** Goldens re-replayed from clean checkouts, provenance labels on every dashboard number, raw episode JSONs committed, resumable runners, and this document’s claim-to-artifact register.

9.3 Where PANOPTES loses — no spin

- **AgentThreatBench memory-poison: 6/10 security on every arm including ours.** The frozen 16-pattern screen does not enroll role-override poison strings, so answer-channel confinement never fires on them; the same three samples (mp_003/004/006) fail on stock, APPA, and PANOPTES arms alike. Their published guarded arm hit security 1.00 at utility 0.50 — on a different model.
- **A ~12-point utility tax on guarded ATB arms.** 0.79 utility for PANOPTES vs 0.92 for the unguarded arm on the same run — recovered security is paid in task completion.
- **Corp utility 13/20 vs 19/20 unguarded.** Recovery routing — the same tax class their published 15/20 guarded result pays.
- **Every number is self-run.** Zero third-party reproduction; the strongest cross-check is the tau-bench exact agreement with the real binary, which is still our harness.
- **Productization gaps.** No npm/PyPI/OCI packaging, no multi-tenancy, no SIEM streaming, no SOC 2, no formal verification of the lattice, no marketplace of integrations — the incumbent ships batteries for a dozen SaaS surfaces and a Helm chart today.
- **Content-side transform evasions.** 49.95% of transformed secrets and 76.87% of homoglyph screen mutations evade the content detectors; the compensating label channel is why the attacks still fail, but the

detectors themselves are honestly partial.

9.4 Choosing among them

The honest selection guide follows from the tables rather than from preference. If the deployment cannot tolerate non-determinism or un-auditable verdicts (compliance, incident forensics, reproducible pipelines), the LLM judge and vendor auto-modes are out regardless of their accuracy. If the deployment has real memory between sessions — and every serious agent deployment does — the label-only incumbent is structurally exposed to T5, reproduced on its own binary. If the deployment needs the integration ecosystem today — marketplace batteries, Claude Code and kAgent adapters, Helm — OpenAPPA is the mature choice and PANOPTES is the research fork to watch. If the deployment can adopt source now and wants the organ delta — memory taint, canary interdiction, injection quarantine, echo confinement, escrowed authority, fail-closed ratchet, recovery affordances, with every claim behind a committed artifact — PANOPTES is buildable from a clean checkout in minutes, and §11 says exactly where to look.

10 Honest Limits and Where It Loses

This chapter consolidates every known limitation with its status and compensating control. It is written to survive an adversarial read: nothing in it is softened, and the open items are open.

Limit	Evidence	Status / compensating control
ATB memory-poison strings pass the screen on all arms (6/10 security)	ATB run: identical failures mp_003/004/006 on stock, APPA, and PANOPTES	Open. Candidate fix: enroll role-override markers from declared signatures; measured before shipping
~12-point utility tax on guarded ATB arms (0.79 vs 0.92)	ATB matched arms, same run	Structural to enforcement; mitigated by recovery affordances, priced in the README
Corp utility 13/20 vs 19/20 open arm	corp LLM run, matched arms	Recovery-routing failures; same class as their published 15/20 guarded tax
Transformed-secret evasions ~49.95% (base64/hex/rot13/reverse)	red-team campaign, content-only channel	Open, documented in SECURITY.md; rotation + egress rate limits at the adapter; label gates still hold the effects
Screen homoglyph/zero-width evasions ~76.87%	red-team campaign, screen channel	Open, documented; compensating control is the label lattice on gated effects
Semantic paraphrase laundering by the model itself	threat-model argument (O7 is verbatim-bounded)	Inherent to the class; canary coverage + egress review are the documented controls
Attacks that need no tool call	threat-model scope	Out of scope by design — the engine mediates actions
Encoded canary exfiltration	not substring-matchable	Rotation policy + adapter rate limits; encoded-scan is an open engineering item

Limit	Evidence	Status / compensating control
All evidence self-run; zero third-party reproduction	§7 in full	Open; harnesses and raw artifacts are committed precisely to make reproduction cheap
No npm/PyPI/OCI packaging, multi-tenancy, SIEM streaming, SOC 2, or formal lattice proofs	productization backlog	Roadmapped (P1–P5 phases with SLO gates); not claimed
2 ATB egress-control episodes pending (76/78)	aggregate module state	Runner resumable; pending episodes are panoptes-arm controls, not upstream samples
Runtime/host compromise	threat-model assumption	Out of scope — harden the sidecar like any control plane

Table 11 — The gap register: every known limitation, its evidence, and its status.

Two of these deserve the last word because they define the research frontier rather than a backlog item. The **memory-poison gap** is not a PANOPTES-specific failure — it is the measured demonstration that a frozen generic pattern set cannot see adversarial novelty, on any arm in the run; closing it without re-importing non-determinism (the O3 counterfactual-adjudication organ, or declared-signature enrollment) is exactly the kind of problem the architecture was built to increment on. And the **utility tax** is the permanent rent of enforcement in this class — visible in our matched arms, in their published numbers, and in every IFC system since Denning: security that never costs a task is security that never enforces one. The product consequence is honest pricing: the tax is measured, printed, and gating the roadmap's utility-recovery phases, not hidden in a footnote.

11 Evidence Register

Every load-bearing claim in this paper, mapped to its committed artifact. All artifacts live in the PANOPTES repository; the JSONs are the raw evidence and the tests re-derive the goldens from a clean checkout.

Claim	Artifact
24/24 attacks + 16/16 benign (organs on); 16/24 + 16/16 (organs off); 20/24 + 12/16 (blunt)	public/bench/panoptes_prototype_benchmarks.json; panoptes bench goldens; tests/regression.test.ts
0.62 µs mean / 0.43 µs organs-off / 615,270 dec/s sustained / p99 3.44 µs / flat heap	prototype-benchmarks latency block; panoptes_extreme_stress_campaign.json
Real binary 16/24 + 16/16 at ~2.08 ms/step; superlinear replay scaling	prototype-benchmarks (openappa arm, scaling probes)
LLM judge 24/24 + 16/16 at ~782 ms, non-deterministic	prototype-benchmarks (llmJudge arm + caveat)
Corp replay 20/20 + 20/20; open 0/20	external_benchmarks.json (corpPolicyReplay)
Corp LLM: open 19/20 + 6/20; APPA 14/20 + 2/20 (laundering); PANOPTES 13/20 + 0/20	external_benchmarks.json (corpLlm, 80 episodes)
Delegation laundering found + closed; regression triplet	corp LLM episodes; panoptes/tests/v2.test.ts

Claim	Artifact
tau 0/853 golden + 64/64 chaos; real-binary exact agreement; 1.08 µs mean	external_benchmarks.json (taubench)
ATB 76/78; stock 0.92/0.77, APPA 0.81/0.81, PANOPTES 0.79/0.79; mp 6/10 all arms	external_benchmarks.json (agentThreatBench, 76 episodes + pending flag)
Red-team: 0/1M label-gated, 0/1M literal, 49.95% transform, 76.87% screen, 0/500K benign FPs	panoptes_redteam_scale.json
149 tests / 54,792 assertions / 12 files; per-file inventory	bun test (panoptes/); Table 6
F-1 decimal evasion found + fixed + regression-tested	panoptes/docs/AUDIT.md; tests/hardening.test.ts
96-bit composite keys property-tested over 200K inputs	tests/hardening-audit.test.ts
Recovery affordances at −9.9% soak throughput	CHANGELOG v1.1.0 entry (soak comparison)
Their published: OpenAPPA 89%/0% (1,320 evals), Claude Auto 90%/10%, FIDES 41%/31%	their GitHub README (project-page numbers)
Their published: corp appa 15/20 + 0/20, open 19/20 + 5/20; ATB Sonnet-5 arms	their results pages (theirPublished block in external_benchmarks.json)
Engine hot paths: 1.04–1.20 µs algebra / 2.65–4.3 µs stack / 4.05–6.0 µs echo	external_benchmarks.json (engine block); fresh re-run 2026-10-21

Table 12 — Claim-to-artifact register (the reproducibility contract of this paper).

The register is also the invitation. Every artifact above is committed; the harnesses are in *scripts/bench2/* with resumable runners; the engine re-derives its goldens from a clean checkout with two commands. Independent reproduction — the single largest gap in the evidence base, stated in §1, §7, and §10 — is one *git clone* and one *bun test* away, and the strongest thing that could happen to these numbers is someone hostile running them.

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